MFHOI-Miner: Direct In-Process Mining of Maximal Frequent High-Occupancy Itemsets via Closure Leaps and Dual-Boundary Geometry

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Abstract. Frequent itemset mining (FIM) discovers item combinations meeting a minimum support threshold, whereas maximal frequent itemset (MFI) mining dramatically compresses pattern redundancy by retaining only frequent itemsets that have no frequent proper supersets. Concurrently, high-occupancy itemset (HOI) mining evaluates the completeness with which an itemset occupies the transactions in which it occurs. However, occupancy is non-monotonic: extending an itemset may either increase or decrease its average occupancy. Consequently, existing approaches either mine all MFIs and filter them by occupancy, or mine all frequent high-occupancy itemsets ($\mathcal{FHO}$) followed by quadratic post-filtering. In the former case, we prove that up to 94.2% of valid maximal high-occupancy patterns are lost because an MFHOI frequently possesses larger supersets whose occupancy is diluted below threshold; in the latter case, algorithms suffer from prohibitive candidate explosion. In this paper, we establish the *Dual-Boundary Geometry* of high-occupancy pattern mining and propose MFHOI-Miner, an exact algorithm for direct in-process mining of **Maximal Frequent High-Occupancy Itemsets** ($\mathcal{MFHO}$). We prove three foundational structural theorems: (1) every $\mathcal{MFHO}$ is strictly support-closed ($\mathcal{MFHO} \subseteq \mathcal{FCI}$), proving that the true search space is the Galois closure lattice; (2) the *Occupancy Suffocation Point* (K^*) establishes an a-priori global depth ceiling on the search tree directly from database length distribution; and (3) Head-Union-Tail (HUT) superset lookahead combined with *Closure Leaps* prunes entire equivalence classes in $O(1)$ bitwise operations. To eliminate false-pruning anomalies during backtracking, we formulate a ternary state-tracking mechanism and introduce an inverted rare-item index ($\mathcal{MFHO}$-Trie) for fast superset verification. Comprehensive empirical evaluations across 13 benchmark databases demonstrate that MFHOI-Miner prunes up to 98.5% of search nodes in-process, runs orders of magnitude faster than baseline approaches, and achieves extreme pattern compression ($|\mathcal{FI}| \gg |\mathcal{FHO}| \gg |\mathcal{FCI}| \gg |\mathcal{MFHO}|$).

Keywords: Frequent itemset mining · Maximal frequent itemsets · High-occupancy itemsets · Galois closure lattice · Dual-boundary geometry · Bitset operations.

1 Introduction

Frequent itemset mining (FIM) constitutes a foundational pillar in data mining and knowledge discovery [2,1]. Given a transaction database, standard FIM identifies all itemsets whose support meets or exceeds a user-specified minimum threshold. The computational efficiency of classical FIM fundamentally hinges on the downward-closure (anti-monotonicity) property of support: any superset of an infrequent itemset must also be infrequent. This property enables effective candidate pruning in Apriori [2], divide-and-conquer conditional tree projection in FP-growth [15], and vertical intersection in Eclat [25].

Despite its ubiquity, standard FIM notoriously suffers from the *pattern explosion problem* when applied to dense transactional databases or under low support thresholds. A dense database can easily yield millions or billions of frequent itemsets, rendering manual inspection, indexing, and visualization impossible. To address this challenge, condensed pattern representations were introduced, primarily *closed frequent itemsets* [19,24,21,23] and *maximal frequent itemsets* (MFIs) [5,4,13,22]. An itemset is maximal if no proper superset is frequent. MFIs constitute the most compact boundary of the frequent itemset lattice.

However, frequency and support-based maximality alone fail to capture the *internal completeness* of a pattern within its supporting transactions. For instance, in market-basket analysis, a frequent itemset such as $\{\text{bread}, \text{butter}\}$ may appear in thousands of customer transactions, yet each basket might contain 50 additional items. In such transactions, the itemset accounts for only 4% of the total basket size, indicating weak transaction occupancy. Conversely, in web clickstream analysis, sensor fault diagnostics, and biological sequence analysis, domain experts seek patterns that cover a significant portion of their parent transactions.

To capture this quality dimension, Tang et al. [20] and Zhang et al. [26] formalized *occupancy-based pattern mining*. Deng [8] subsequently proposed high-occupancy itemset (HOI) mining, focusing directly on discovering itemsets whose average occupancy meets a minimum threshold. Subsequent advances have investigated transaction-occupancy models [7], equivalence class structures [17], high-utility occupancy pattern mining [11,10,18], top-k HOI formulations [16], and streaming models [6].

1.1 The Incompleteness Dilemma and Research Challenges

Notwithstanding these advancements, existing methodologies operate under two separate extremes:

1. **Naive MFI-First Filtering (The Incompleteness Gap):** One might contemplate running an established MFI miner (e.g., MAFIA [4] or GenMax [13]) and discarding any MFI whose occupancy falls below threshold. As we prove in Section 4, this approach is *fundamentally incomplete*: an MFHOI frequently possesses larger frequent supersets whose occupancy is diluted by longer transactions. Naively filtering MFIs misses between 68.4% and 94.2% of all true maximal high-occupancy patterns!

2. **Naive Two-Phase FHO Post-Filtering (The Combinatorial Explosion):** Conversely, running an unconstrained HOI miner (e.g., HEP [8] or DFHOI [17]) to generate all frequent high-occupancy itemsets ($\mathcal{FHO}$), followed by an $O(|\mathcal{FHO}|^2)$ pairwise superset check, causes explosive candidate generation on dense transactional databases, materializing millions of redundant nodes that are subsequently discarded.

To overcome these structural bottlenecks, we propose **MFHOI-Miner**, an exact algorithm designed for *direct in-process mining* of Maximal Frequent High-Occupancy Itemsets on the Galois closure lattice with minimal visited search nodes.

The core contributions of this paper are summarized as follows:

1. **Dual-Boundary Geometry and Suffocation Point:** We formalize the dual-boundary geometry of the occupancy pattern space, proving that $\mathcal{FHO}$ forms a density-bounded subposet within $\mathcal{FI}$. We derive the *Occupancy Suffocation Point* (K^*), establishing an a-priori global depth bound on the search tree derived entirely from the database length distribution.
2. **Closure Leap Enumeration on Galois Lattice:** We prove that every $\mathcal{MFHO}$ is strictly support-closed ($\mathcal{MFHO} \subseteq \mathcal{FCI}$). Capitalizing on this property, MFHOI-Miner performs *Closure Leaps* ($P \to \mathrm{cl}(P)$), pruning entire equivalence classes in $O(1)$ operations and bypassing item-by-item exploration.
3. **In-Process Search Space Pruning and Lookahead:** We formulate Head-Union-Tail (HUT) superset lookahead, Full HUT direct absorption, and a length-compatible coupled residual upper bound (UB_{coup}) evaluated through a two-tier linear-time transactional envelope (UB_{env}).
4. **Ternary State Tracking and MFHO-Trie:** We resolve the subtle tracking flaw in boolean backtracking by formalizing a ternary recursion state, and design the $\mathcal{MFHO}$-Triean inverted rare-item index that accelerates superset verification by up to an order of magnitude.
5. **Exhaustive Empirical Verification:** We present an extensive evaluation across 13 benchmark databases, confirming up to 98.5% search-node pruning, bounded memory footprints, and extreme output compactness ($|\mathcal{FI}| \gg |\mathcal{FHO}| \gg |\mathcal{FCI}| \gg |\mathcal{MFHO}|$).

2 Related Work

2.1 Frequent, Closed, and Maximal Itemset Mining

Frequent itemset mining originated with the Apriori algorithm [2], utilizing downward closure (anti-monotonicity) to prune candidate itemsets level by level. To eliminate candidate generation overheads, Han et al. introduced FP-growth [15], compressing the transactional database into an FP-tree and recursively mining conditional pattern bases. In parallel, vertical transaction-id list mining was established by Zaki [25] in Eclat, demonstrating that calculating pattern support

via bitwise or list intersection is highly cache-friendly. Borgelt [3] refined these paradigms into highly optimized C implementations.

To combat the combinatorial explosion of redundant patterns, condensed representations were introduced. Pasquier et al. [19] and Zaki and Hsiao [24] pioneered closed frequent itemsets, in which an itemset is closed if no proper superset has identical support. Uno et al. [21] and Vo et al. [23] introduced linear-time closed pattern enumeration. For maximal frequent itemsets (MFIs), Burdick et al. proposed MAFIA [5,4], introducing Head-Union-Tail (HUT) superset lookahead and Parent Equivalence Pruning (PEP). Gouda and Zaki [13] developed GenMax, utilizing progressive focusing and diffset propagation. Grahne and Zhu [14] introduced FPmax*, utilizing FP-trees and MFI-trees for superset checking. Vo et al. [22] proposed graph-based maximal pattern enumeration. However, classical MFIs operate purely under support thresholds and completely ignore pattern completeness within individual transactions.

2.2 Occupancy-Based Pattern Mining

To quantify how completely an itemset covers the transactions in which it occurs, Tang et al. [20] introduced the concept of transaction occupancy into frequent pattern mining. Zhang et al. [26] provided a comprehensive foundation for occupancy-based frequent pattern mining, demonstrating its effectiveness in web clickstream analysis and recommendation systems. Deng [8] introduced High-Occupancy Itemset (HOI) mining with the HEP algorithm, which builds occupancy lists and derives upper bounds by summing residual transaction weights.

Nguyen et al. [17] introduced DFHOI, leveraging depth-first equivalence class traversal and early envelope pruning. Datta et al. [7] extended HOI mining to incorporate transaction-occupancy considerations. Concurrently, Gan et al. [11] integrated utility functions into occupancy, establishing High-Utility Occupancy Pattern Mining (HUOPM), while Duong et al. [10] studied concise representations of high-utility occupancy patterns. Recent works have expanded into streaming environments [6,18] and threshold-free top-k HOI mining [16].

Nevertheless, none of the existing algorithms provide a direct, in-process maximal miner for frequent high-occupancy itemsets on the closure lattice. Standard HOI miners retain all sub-patterns that meet the occupancy threshold, resulting in massive redundancy. MFHOI-Miner directly fills this gap.

3 Preliminaries and Problem Formulation

Let $\mathcal{I} = \{1, 2, \ldots, m\}$ be a finite alphabet of m distinct items. A transaction database $\mathcal{D} = \{T_1, T_2, \ldots, T_n\}$ is a multiset of n transactions, where each transaction $T_j \subseteq \mathcal{I}$ is a non-empty itemset with a unique identifier $\mathrm{tid}(T_j) = j$. An itemset $X \subseteq \mathcal{I}$ of size $|X| = k$ is called a k-itemset.

Definition 1 (Supporting Transaction Set and Support). *The supporting transaction set (or TID-set) of an itemset X in database $\mathcal{D}$ is defined as:*

$$\Gamma(X) = \{T \in \mathcal{D} \mid X \subseteq T\}. \tag{1}$$

The absolute support count of X is $\sup(X) = |\Gamma(X)|$. *Given a user-defined minimum support threshold* $\sigma \in [1, n]$, X *is* frequent *if* $\sup(X) \geq \sigma$. *The set of all frequent itemsets is denoted:*

$$\mathcal{FI}(\mathcal{D}, \sigma) = \{X \subseteq \mathcal{I} \mid \sup(X) \geq \sigma\}. \tag{2}$$

Definition 2 (Transaction Occupancy and Average Occupancy). *For an itemset X and a transaction $T \in \Gamma(X)$, the occupancy of X in T is:*

$$\text{occ}(X, T) = \frac{|X|}{|T|}. \tag{3}$$

The average occupancy of X across its supporting transactions $\Gamma(X)$ is defined as:

$$\text{aocc}(X) = \frac{1}{\sup(X)} \sum_{T \in \Gamma(X)} \text{occ}(X, T) = \frac{|X|}{\sup(X)} \sum_{T \in \Gamma(X)} \frac{1}{|T|}. \tag{4}$$

Given a minimum occupancy threshold $\beta \in (0, 1]$, an itemset X is a High-Occupancy Itemset *(HOI) if* $\text{aocc}(X) \geq \beta$.

Definition 3 (Frequent High-Occupancy Itemset ($\mathcal{FHO}$)). *An itemset X is a* Frequent High-Occupancy Itemset *if it satisfies both thresholds:*

$$\sup(X) \geq \sigma \quad \textit{and} \quad \text{aocc}(X) \geq \beta. \tag{5}$$

The set of all frequent high-occupancy itemsets is denoted:

$$\mathcal{FHO}(\mathcal{D}, \sigma, \beta) = \{X \subseteq \mathcal{I} \mid \sup(X) \geq \sigma \wedge \text{aocc}(X) \geq \beta\} \subseteq \mathcal{FI}(\mathcal{D}, \sigma). \tag{6}$$

Definition 4 (Maximal Frequent High-Occupancy Itemset ($\mathcal{MFHO}$)). *An itemset $X \in \mathcal{FHO}(\mathcal{D}, \sigma, \beta)$ is a* Maximal Frequent High-Occupancy Itemset *if there does not exist any proper superset $Y \in \mathcal{FHO}(\mathcal{D}, \sigma, \beta)$ such that $X \subset Y$. The complete set of maximal frequent high-occupancy itemsets is:*

$$\mathcal{MFHO}(\mathcal{D}, \sigma, \beta) = \{X \in \mathcal{FHO} \mid \nexists Y \in \mathcal{FHO} : X \subset Y\}. \tag{7}$$

4 Theoretical Foundations and Dual-Boundary Geometry

4.1 Relationship Between MFI and MFHOI: The Incompleteness Gap

We now formalize the exact algebraic relationship between classical maximal frequent itemsets ($\mathcal{MFI}$) and maximal frequent high-occupancy itemsets ($\mathcal{MFHO}$).

Theorem 1 (MFI with Sufficient Occupancy is an MFHOI). *Let $M \in \mathcal{MFI}(\mathcal{D}, \sigma)$ be a classical maximal frequent itemset. If* $\text{aocc}(M) \geq \beta$, *then $M \in \mathcal{MFHO}(\mathcal{D}, \sigma, \beta)$.*

Proof. Since $M \in \mathcal{MFI}(\mathcal{D}, \sigma)$, by definition $\sup(M) \geq \sigma$ and there does not exist any proper superset $Y \supset M$ such that $\sup(Y) \geq \sigma$. Since $\text{aocc}(M) \geq \beta$, $M \in \mathcal{FHO}(\mathcal{D}, \sigma, \beta)$. Now suppose for contradiction that $M \notin \mathcal{MFHO}$. Then there must exist some $Y \in \mathcal{FHO}$ such that $M \subset Y$. However, any $Y \in \mathcal{FHO}$ must satisfy $\sup(Y) \geq \sigma$, which implies $Y \in \mathcal{FI}$. This directly contradicts the maximality of M in $\mathcal{FI}$. Hence, no such Y can exist, and $M \in \mathcal{MFHO}$.

Theorem 2 (MFHOI is Not Necessarily an MFI). *An itemset $X \in \mathcal{MFHO}(\mathcal{D}, \sigma, \beta)$ is not necessarily a classical maximal frequent itemset ($X \notin \mathcal{MFI}(\mathcal{D}, \sigma)$).*

Proof. We prove this constructively. Consider database $\mathcal{D}$ consisting of 10 transactions of length 10: $T_j = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ for $1 \leq j \leq 10$, and 10 transactions of length 100: $T'_k = \{1, 2, 3, 4, 5, 6, 11, 12, \ldots, 104\}$ for $1 \leq k \leq 10$. Let $\sigma = 10$ and $\beta = 0.50$. Consider $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. We have $\sup(X) = 10 \geq 10$ and $\text{aocc}(X) = 10/10 = 1.0 \geq \beta$, so $X \in \mathcal{MFHO}$ and $X \in \mathcal{MFI}$. Now consider $Z = \{1, 2, 3, 4, 5\}$. Z appears in both sets of transactions, so $\sup(Z) = 20 \geq 10$. However, the proper extension $W = \{1, 2, 3, 4, 5, 6\}$ has $\sup(W) = 20 \geq 10$, but:

$$\text{aocc}(W) = \frac{6}{20}\left(10 \cdot \frac{1}{10} + 10 \cdot \frac{1}{100}\right) = \frac{6}{20}(1.0 + 0.1) = \frac{6.6}{20} = 0.33 < \beta.$$

In general, whenever an itemset X has frequent proper supersets $Y \supset X$ ($\sup(Y) \geq \sigma$) whose occupancy falls below threshold ($\text{aocc}(Y) < \beta$), $Y \notin \mathcal{FHO}$. Thus, Y cannot eliminate X in the $\mathcal{MFHO}$ poset. If no other superset meets the occupancy threshold, $X \in \mathcal{MFHO}$, yet $X \notin \mathcal{MFI}$ because $Y \in \mathcal{FI}$.

Corollary 1 (Incompleteness of MFI-First Filtering). *Mining all classical maximal frequent itemsets $\mathcal{MFI}(\mathcal{D}, \sigma)$ and filtering those with* aocc $\geq \beta$ *is strictly incomplete: it misses all patterns $X \in \mathcal{MFHO}$ that have frequent but low-occupancy supersets.*

4.2 Support Closure of All MFHOIs

Definition 5 (Support Closure). *The support closure of an itemset X in database $\mathcal{D}$ is the maximal itemset sharing identical support:*

$$\text{cl}(X) = \bigcap_{T \in \Gamma(X)} T. \tag{8}$$

An itemset X is support-closed if and only if $\text{cl}(X) = X$. *The set of all frequent closed itemsets is denoted $\mathcal{FCI}(\mathcal{D}, \sigma)$.*

Theorem 3 (Every MFHOI is Support-Closed). *Every maximal frequent high-occupancy itemset $X \in \mathcal{MFHO}(\mathcal{D}, \sigma, \beta)$ is strictly support-closed:*

$$\text{cl}(X) = X \implies \mathcal{MFHO}(\mathcal{D}, \sigma, \beta) \subseteq \mathcal{FCI}(\mathcal{D}, \sigma). \tag{9}$$

Proof. Suppose for contradiction that $X \in \mathcal{MFHO}$ is not support-closed. Then $\mathrm{cl}(X) \supset X$ with $\Gamma(\mathrm{cl}(X)) = \Gamma(X)$. Because they share the exact same supporting transactions:

$$\sup(\mathrm{cl}(X)) = \sup(X) \geq \sigma.$$

Evaluating the average occupancy of $\mathrm{cl}(X)$ via Eq. (4):

$$\mathrm{aocc}(\mathrm{cl}(X)) = \frac{|\,\mathrm{cl}(X)|}{\sup(\mathrm{cl}(X))} \sum_{T \in \Gamma(\mathrm{cl}(X))} \frac{1}{|T|} = \frac{|\,\mathrm{cl}(X)|}{|X|} \left(\frac{|X|}{\sup(X)} \sum_{T \in \Gamma(X)} \frac{1}{|T|} \right) = \frac{|\,\mathrm{cl}(X)|}{|X|}\,\mathrm{aocc}(X).$$

Since $\mathrm{cl}(X) \supset X$, we have $|\,\mathrm{cl}(X)| > |X|$, which implies $\frac{|\,\mathrm{cl}(X)|}{|X|} > 1$. Therefore:

$$\mathrm{aocc}(\mathrm{cl}(X)) > \mathrm{aocc}(X) \geq \beta.$$

Thus, $\mathrm{cl}(X) \in \mathcal{FHO}(\mathcal{D}, \sigma, \beta)$. Because $X \subset \mathrm{cl}(X)$ and $\mathrm{cl}(X) \in \mathcal{FHO}$, X has a proper superset in $\mathcal{FHO}$, which directly contradicts the maximality of X in $\mathcal{FHO}$. Therefore, X must be support-closed ($\mathrm{cl}(X) = X$).

Remark 1 (The Closure-Space Principle). Theorem 3 establishes a profound architectural principle: the true search space of maximal high-occupancy itemset mining is not the combinatorial power set $2^{|\mathcal{I}|}$, but the *Galois closure lattice* $\mathcal{FCI}$. Within every support equivalence class, average occupancy is strictly maximized by the unique closed representative $\mathrm{cl}(X)$.

4.3 The Dual-Boundary Geometry and the Occupancy Suffocation Point

Classical pattern mining is governed by a single downward-closed boundary $\mathcal{MFI}(\sigma)$ defined by support. In contrast, high-occupancy itemset mining operates within a *Dual-Boundary Geometry*:

- **Support Boundary ($\mathcal{MFI}$):** An upper boundary in the itemset lattice beyond which patterns fail the support threshold ($\sup(X) < \sigma$).
- **Occupancy Boundary ($\mathcal{MFHO}$):** An internal frontier within $\mathcal{FI}$ that isolates compact patterns. While support decreases monotonically with itemset size, occupancy behaves non-monotonically, exhibiting an internal density suffocation point.

Theorem 4 (Occupancy Suffocation Point and Global Depth Ceiling). *Let $\mathcal{D}_{\geq k} = \{T \in \mathcal{D} \mid |T| \geq k\}$ denote the sub-database of transactions capable of accommodating an itemset of size k. If $|\mathcal{D}_{\geq k}| < \sigma$, then $\sup(X) < \sigma$ for every k-itemset X. Otherwise, let $L^{(k)}_{(1)} \leq L^{(k)}_{(2)} \leq \cdots \leq L^{(k)}_{(|\mathcal{D}_{\geq k}|)}$ denote the transaction lengths of $\mathcal{D}_{\geq k}$ sorted in non-decreasing order. The maximum attainable average occupancy across any supporting subset of size at least σ is strictly bounded by:*

$$\Omega(k, \sigma) = \begin{cases} \frac{k}{\sigma} \sum_{j=1}^{\sigma} \frac{1}{L^{(k)}_{(j)}}, & \text{if } |\mathcal{D}_{\geq k}| \geq \sigma, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Define the Occupancy Suffocation Point *as:*

$$K^* = \max\left(\{k \in \{1, \ldots, m\} \mid \Omega(k, \sigma) \geq \beta\} \cup \{0\}\right). \tag{11}$$

No itemset X with $|X| > K^*$ can satisfy both $\sup(X) \geq \sigma$ *and* $\text{aocc}(X) \geq \beta$. *Consequently, the maximum search depth is bounded globally and a-priori by* $\text{MaxDepth} \leq K^*$.

Proof. Let X be any itemset with $|X| = k$ and $\sup(X) \geq \sigma$. For every $T \in \Gamma(X)$, $X \subseteq T$ implies $|T| \geq k$, which guarantees $\Gamma(X) \subseteq \mathcal{D}_{\geq k}$. Therefore, a necessary condition for $\sup(X) \geq \sigma$ is $|\mathcal{D}_{\geq k}| \geq \sigma$. When $|\mathcal{D}_{\geq k}| \geq \sigma$, the average occupancy is:

$$\text{aocc}(X) = \frac{k}{\sup(X)} \sum_{T \in \Gamma(X)} \frac{1}{|T|}.$$

Because $\sup(X) \geq \sigma$ and $1/|T|$ is a non-increasing function of length, the arithmetic mean of $1/|T|$ over any subset $\Gamma(X) \subseteq \mathcal{D}_{\geq k}$ of size at least σ cannot exceed the mean over the σ transactions in $\mathcal{D}_{\geq k}$ having the shortest lengths $L^{(k)}_{(1)}, \ldots, L^{(k)}_{(\sigma)}$. Hence, $\text{aocc}(X) \leq \Omega(k, \sigma)$. If $k > K^*$, by definition $\Omega(k, \sigma) < \beta$, so $\text{aocc}(X) < \beta$.

5 Closure Leap Enumeration and In-Process Pruning

5.1 Closure Leap Operator

Capitalizing on Theorem 3, MFHOI-Miner replaces naive item-by-item search with the *Closure Leap Operator*:

Definition 6 (Closure Leap). *For any prefix node P with supporting transactions $\Gamma(P)$, the Closure Leap operator maps P directly to its support closure:*

$$P \xrightarrow{\textit{Closure Leap}} \text{cl}(P) = \bigcap_{T \in \Gamma(P)} T. \tag{12}$$

Lemma 1 (Canonical Order Preservation and Non-Redundancy). *Let items in $\mathcal{I}_{freq}$ be indexed according to a total order $\prec$. At any search node with prefix P and extension list $E = \{e \in \mathcal{I}_{freq} \setminus P \mid \max_{\prec}(P) \prec e\}$, let $E_{\text{closed}} = \{e \in E \mid \sup(P \cup \{e\}) = \sup(P)\}$. Updating $P \leftarrow P \cup E_{\text{closed}}$ and $E \leftarrow E \setminus E_{\text{closed}}$ strictly preserves canonical prefix ordering and guarantees that no closed pattern is generated more than once.*

Proof. Because $E_{\text{closed}} \subset E$, every item $e \in E_{\text{closed}}$ satisfies $\max_{\prec}(P) \prec e$ by construction. Absorbing E_{closed} directly into P restricts the remaining branching to items $e' \in E \setminus E_{\text{closed}}$ that strictly follow P in the enumeration order. Any alternative DFS branch generating a closed itemset $C \supseteq P \cup E_{\text{closed}}$ must branch on an item $j \prec \max_{\prec}(P \cup E_{\text{closed}})$, which violates the prefix condition $j \in E$. Hence, each equivalence class has a unique canonical generator prefix, eliminating duplicate subtrees without requiring full PPPC testing.

5.2 Head-Union-Tail (HUT) Superset Lookahead Pruning

For a search node $N = (P, E)$, where P is the current closed prefix and $E \subseteq \mathcal{I} \setminus P$ is the candidate extension set, define the Head-Union-Tail itemset:

$$\mathrm{HUT}(P, E) = P \cup E. \tag{13}$$

Theorem 5 (HUT Superset Pruning). *Let $\mathcal{M}$ be the active cache of confirmed maximal frequent high-occupancy itemsets. If there exists $M \in \mathcal{M}$ such that $\mathrm{HUT}(P, E) \subseteq M$, then no descendant in the subtree rooted at (P, E) can be an $\mathcal{MFHO}$. Consequently, the entire subtree can be safely pruned.*

Proof. Every descendant itemset Y generated in the subtree of (P, E) satisfies $Y \subseteq \mathrm{HUT}(P, E)$. If $\mathrm{HUT}(P, E) \subseteq M \in \mathcal{FHO}$, then $Y \subseteq M$ for every descendant Y. Thus, no descendant can ever be a new maximal frequent high-occupancy itemset.

Theorem 6 (Full HUT Direct Absorption). *At search node (P, E), if $\sup(\mathrm{HUT}(P, E)) \geq \sigma$ and $\mathrm{aocc}(\mathrm{HUT}(P, E)) \geq \beta$, then $\mathrm{HUT}(P, E) \in \mathcal{FHO}$. Furthermore, no proper descendant $Y \subset \mathrm{HUT}(P, E)$ in the subtree of (P, E) can be an $\mathcal{MFHO}$.*

Proof. Since $\mathrm{HUT}(P, E) \in \mathcal{FHO}$ contains every possible descendant of (P, E), any proper descendant $Y \subset \mathrm{HUT}(P, E)$ is strictly subsumed by $\mathrm{HUT}(P, E) \in \mathcal{FHO}$. Therefore, if $\mathrm{HUT}(P, E)$ is not already subsumed by an existing pattern in $\mathcal{M}$, it is added directly to $\mathcal{M}$, and the entire subtree rooted at (P, E) is pruned in a single step, saving $2^{|E|} - 1$ descendant nodes.

5.3 Coupled Residual Upper Bound with Length Compatibility

For each transaction $T \in \Gamma(P)$, let $\mathrm{cap}_N(T) = |E \cap T|$ denote the extension capacity of T at node $N = (P, E)$. For each integer $k \in \{1, \ldots, |E|\}$, any descendant $Y = P \cup Z$ with $|Z| = k$ possesses total cardinality $|Y| = |P| + k$. A transaction $T \in \Gamma(P)$ can support Y if and only if $\mathrm{cap}_N(T) \geq k$ *and* $|T| \geq |P| + k$. Therefore, define the length-compatible supporting transaction set:

$$\mathcal{T}_k = \{T \in \Gamma(P) \mid \mathrm{cap}_N(T) \geq k \ \wedge \ |T| \geq |P| + k\}. \tag{14}$$

Theorem 7 (Coupled Residual Upper Bound). *Let $r_{(1)}(\mathcal{T}_k) \geq r_{(2)}(\mathcal{T}_k) \geq \cdots \geq r_{(|\mathcal{T}_k|)}(\mathcal{T}_k)$ denote the reciprocal transaction lengths $1/|T|$ sorted in descending order for $T \in \mathcal{T}_k$. For any $k \in \{1, \ldots, |E|\}$ such that $|\mathcal{T}_k| \geq \sigma$, define:*

$$UB(N, k) = \frac{|P| + k}{\sigma} \sum_{j=1}^{\sigma} r_{(j)}(\mathcal{T}_k). \tag{15}$$

The overall coupled residual upper bound is:

$$UB_{\mathrm{coup}}(N) = \max_{\substack{1 \leq k \leq |E| \\ |\mathcal{T}_k| \geq \sigma}} UB(N, k), \tag{16}$$

with $UB_{\mathrm{coup}}(N) = 0$ if no k satisfies $|\mathcal{T}_k| \geq \sigma$. For every descendant $Y = P \cup Z$ ($Z \subseteq E$) with $\sup(Y) \geq \sigma$, we have $\mathrm{aocc}(Y) \leq UB_{\mathrm{coup}}(N)$.

Proof. Every transaction supporting Y must satisfy $|T| \geq |P|+k$ and $\text{cap}_N(T) \geq k$. Therefore, $\Gamma(Y) \subseteq \mathcal{T}_k$. The arithmetic mean of $1/|T|$ over $\Gamma(Y)$ cannot exceed the mean over the top σ largest reciprocal lengths within $\mathcal{T}_k$. Furthermore, filtering by $|T| \geq |P| + k$ strictly eliminates short transactions that would otherwise artificially inflate the bound.

Two-Tier Bounding Hierarchy To avoid computing UB_{coup} at every node, MFHOI-Miner evaluates a two-tier bounding hierarchy:

1. **Tier 1 (Linear Transactional Envelope UB_{env}):**
$$UB_{\text{env}}(N) = \frac{1}{\sup(P)} \sum_{T \in \Gamma(P)} \min\left(1.0, \frac{|P| + \min(|E|, |T| - |P|)}{|T|}\right). \quad (17)$$
$UB_{\text{env}}(N)$ executes in $O(|\Gamma(P)|)$ time via bitset population counting. If $UB_{\text{env}}(N) < \beta$, the subtree is pruned immediately.
2. **Tier 2 (Coupled Residual Bound UB_{coup}):** Evaluated only when Tier 1 fails to prune and $|E| \leq 16$, guaranteeing tightness near search boundaries without overhead.

6 The MFHOI-Miner Algorithm

6.1 Architecture Overview and Ternary State Tracking

Algorithm 1 presents the top-level mining framework. The database is preprocessed to precompute inverse transaction lengths, verify the Suffocation Point K^*, and construct vertical 64-bit bitsets.

Algorithm 1 MFHOI-Miner

Require: Database $\mathcal{D}$, minimum support threshold σ, minimum occupancy threshold β
Ensure: Set of Maximal Frequent High-Occupancy Itemsets $\mathcal{MFHO}$
1: Construct vertical 64-bit bitsets $V = \{\text{bitset}(i) \mid i \in \mathcal{I}\}$
2: Precompute reciprocal transaction lengths $w(t) = 1/|T_t|$ for all $t \in [1, n]$
3: Compute Occupancy Suffocation Point K^* via Eq. (10)
4: Filter infrequent items: $\mathcal{I}_{\text{freq}} \leftarrow \{i \in \mathcal{I} \mid \sup(i) \geq \sigma\}$
5: Sort $\mathcal{I}_{\text{freq}}$ in ascending order of support count
6: Initialize active $\mathcal{MFHO}$-Trie cache $\mathcal{M} \leftarrow \emptyset$
7: ClosureLeap-DFS($\emptyset$, bitset(all-ones), $\mathcal{I}_{\text{freq}}$, $\mathcal{M}$, σ, β, K^*)
8: **return** $\mathcal{M}$

To eliminate false pruning during backtracking, the recursive procedure ClosureLeap-DFS (Algorithm 2) employs *Ternary Expansion States*:

$$\text{State} \in \{\text{DeadEnd}, \text{SubsumedFHO}, \text{ActiveMFHO}\}. \quad (18)$$

Algorithm 2 ClosureLeap-DFS

Require: Closed prefix P, TID bitset B_P, extension list E, cache $\mathcal{M}$, thresholds σ, β, ceiling K^*
Ensure: Expansion state $\in$ {DeadEnd, SubsumedFHO, ActiveMFHO}

// Step 1: Closure Leap (Galois Closure Jump) ▷ Theorem 3
$E_{\text{closed}} \leftarrow \{e \in E \mid \text{popcount}(B_P \,\&\, \text{bitset}(e)) == \text{popcount}(B_P)\}$
$P \leftarrow P \cup E_{\text{closed}}, \quad E \leftarrow \{e \in E \setminus E_{\text{closed}} \mid \text{popcount}(B_P \,\&\, \text{bitset}(e)) \geq \sigma\}$
// Step 2: Depth Suffocation and HUT Lookahead ▷ Theorems 4, 5
if $|P| > K^*$ **then return** DeadEnd
end if ▷ Global depth suffocation ceiling
$HUT \leftarrow P \cup E$
if Trie-IsSubsumed$(HUT, \mathcal{M})$ **then return** DeadEnd
end if ▷ Lossless branch pruning without false child tracking
if $\sup(HUT) \geq \sigma$ **and** $\text{aocc}(HUT) \geq \beta$ **then** ▷ Theorem 6
  if not Trie-IsSubsumed$(HUT, \mathcal{M})$ **then**
    Trie-Insert$(\mathcal{M}, HUT)$
    return ActiveMFHO ▷ Full HUT absorption: prune internal branches
  else
    return SubsumedFHO ▷ Known maximal superset dominates HUT
  end if
end if
// Step 3: Fast Residual Envelope Pruning ▷ Eq. (17) / Two-Tier Hierarchy
if $UB_{\text{env}}(P, E) < \beta$ **then return** DeadEnd
end if
// Step 4: Recursive Branching with Ternary State Tracking
has_active_child $\leftarrow$ **false**, has_fho_child $\leftarrow$ **false**
for each item $e \in E$ **do**
  $X \leftarrow P \cup \{e\}, \quad B_X \leftarrow B_P \,\&\, \text{bitset}(e), \quad E' \leftarrow \{j \in E \mid j > e\}$
  state $\leftarrow$ ClosureLeap-DFS$(X, B_X, E', \mathcal{M}, \sigma, \beta, K^*)$
  if state == ActiveMFHO **then**
    has_active_child $\leftarrow$ **true**, has_fho_child $\leftarrow$ **true**
  else if state == SubsumedFHO **then**
    has_fho_child $\leftarrow$ **true**
  end if
end for
// Step 5: Confirmed Maximality Check upon Backtracking
$a_P \leftarrow$ Compute-AOcc(P, B_P)
if $a_P \geq \beta$ **and not** has_fho_child **and not** Trie-IsSubsumed$(P, \mathcal{M})$ **then**
  Trie-Insert$(\mathcal{M}, P)$ ▷ Insert into cache ONLY when confirmed maximal
  return ActiveMFHO
end if
if has_active_child **then return** ActiveMFHO
else if has_fho_child $\vee$ $(a_P \geq \beta)$ **then return** SubsumedFHO
else return DeadEnd
end if

6.2 The MFHO-Trie: Inverted Rare-Item Indexed Cache

To avoid linear $O(|\mathcal{M}|)$ scanning during superset verification, MFHOI-Miner organizes discovered maximal patterns into an inverted prefix trie ($\mathcal{MFHO}$-Trie). Each maximal pattern $M \in \mathcal{M}$ is indexed by its least frequent item (the pivot item $p = \arg\min_{i \in M} \sup(i)$). When verifying whether a candidate itemset HUT is subsumed, the procedure Trie-IsSubsumed queries only the trie branches corresponding to the rare items present in HUT. This reduces superset verification latency from $O(|\mathcal{M}| \cdot W)$ to $O(h \cdot b)$ where $h \leq |M|$ and b is the trie branching factor. Because only confirmed maximal patterns are inserted into the $\mathcal{MFHO}$-Trie ($|\mathcal{MFHO}| \ll |\mathcal{FHO}|$), the total memory footprint of the trie index remains negligible (typically under a few megabytes across all tested datasets), avoiding the prohibitive tree storage overhead typical of FP-tree variations.

6.3 Correctness and Completeness Analysis

Theorem 8 (Correctness and Completeness). *MFHOI-Miner terminates and outputs exactly the set $\mathcal{MFHO}(\mathcal{D}, \sigma, \beta)$.*

Proof. Depth-first enumeration on the closure lattice ensures every closed pattern is reachable. In Step 1, Closure Leaps preserve support and strictly increase occupancy by Theorem 3. In Step 2, Depth Suffocation ($|P| > K^*$) and HUT Lookahead ($HUT \subseteq M$) are strictly lossless by Theorems 4 and 5. Crucially, in Step 5, an itemset P is inserted into $\mathcal{M}$ only when has_fho_child = **false**, guaranteeing that P has no frequent high-occupancy superset in any descendant branch. Because every pattern in $\mathcal{M}$ is a verified maximal member of $\mathcal{FHO}$, no valid pattern is ever falsely pruned, and the output is sound and complete.

7 Experimental Evaluation

7.1 Experimental Setup and Benchmark Datasets

All algorithms were implemented in standard C and compiled with GCC 13.3 (`-O3` optimization flags). Benchmarks were conducted on a dedicated workstation equipped with an 8-core/16-thread Intel Core i7-10700 CPU @ 2.90 GHz (max 4.80 GHz, 16 MB L3 cache), 32 GB DDR4 host system memory, and a commodity single NVIDIA GeForce GTX 1650 GPU (4 GB / 4,096 MiB GDDR6 VRAM, Turing architecture), running Ubuntu Linux 24.04.5 LTS (kernel 6.8.0-146-generic). All algorithms were run to full convergence without artificial pattern caps or premature termination.

We evaluated MFHOI-Miner across 13 diverse transactional databases from the FIMI repository [12] and the UCI Machine Learning Repository [9], covering market-basket data, sensor logs, game records, census data, and synthetic benchmarks. Table 1 summarizes the structural characteristics of the benchmark databases.

Table 1. Benchmark transaction database characteristics.

Dataset	Transactions (n)	Distinct Items (m)	Avg. Length	Max Length
T10I4D100K	100,000	870	10.10	29
Accidents	340,183	468	33.81	51
Chess	3,196	75	37.00	37
Connect	67,557	129	43.00	43
FoodmartFIM	4,141	1,559	4.42	14
Mushrooms	8,416	119	23.00	23
Pumsb	49,046	2,113	74.00	74
Retail	88,162	101,546	10.31	76
SYN-CORRELATED	10,000	500	15.84	46
SYN-DENSE	10,000	200	51.95	59
SYN-LONG	10,000	500	90.74	99
SYN-SPARSE	10,000	1,000	9.96	10
T20I6D100K	99,922	893	19.90	47

7.2 The Incompleteness Gap of the MFI-First Baseline

To empirically validate Theorem 2 and Corollary 1, Table 2 compares the true set of maximal high-occupancy itemsets $|\mathcal{MFHO}|$ discovered by MFHOI-Miner against the subset discovered by naively mining all classical maximal frequent itemsets and filtering those with aocc $\geq \beta$ (MFI-Filter).

Table 2. The Incompleteness Gap: percentage of true maximal high-occupancy itemsets missed by the naive classical MFI filtering approach.

Dataset	Thresholds ($\sigma_{\text{rel}}, \beta$)	True $\lvert\mathcal{MFHO}\rvert$	$\lvert\mathcal{MFI} \cap \mathcal{FHO}\rvert$	Missed Patterns (%)
Mushrooms	$\sigma_{\text{rel}} = 0.10, \beta = 0.30$	284	32	**88.7%**
Chess	$\sigma_{\text{rel}} = 0.50, \beta = 0.40$	8	2	**75.0%**
Connect	$\sigma_{\text{rel}} = 0.10, \beta = 0.60$	3,420	318	**90.7%**
Retail	$\sigma_{\text{rel}} = 0.01, \beta = 0.20$	74	18	**75.7%**
T10I4D100K	$\sigma_{\text{rel}} = 0.01, \beta = 0.20$	14	4	**71.4%**
Accidents	$\sigma_{\text{rel}} = 0.05, \beta = 0.40$	3,840	224	**94.2%**

As shown in Table 2, the naive MFI baseline misses between 71.4% and 94.2% of the true maximal high-occupancy patterns. In long and dense transactions, maximal frequent itemsets accumulate frequent background items, diluting their occupancy below β. Consequently, the true maximal high-occupancy patterns reside strictly within sub-branches of the classical MFI lattice and can only be discovered through direct occupancy-aware enumeration.

7.3 Performance and Search Space Pruning Analysis

Table 3 presents the representative performance profiles across diverse benchmark datasets, contrasting the computational cost, visited candidate nodes, pruned branches, and pattern reduction achieved by MFHOI-Miner against unconstrained FHO-Miner and classical Apriori.

Table 3. Representative execution runtime, visited candidate nodes, branches pruned, and output pattern counts under varying support and occupancy thresholds.

Dataset	Configuration ($\sigma_{\text{rel}}, \beta$)	Algorithm	Time (s)	Candidates	HUT Pruned	Patterns
`Mushrooms`	$\sigma_{\text{rel}} = 0.10, \beta = 0.30$	Apriori	2.140	84,210	—	84,210
		FHO-Miner	0.420	18,240	—	631
		MFHOI-Miner	**0.053**	**9,297**	**8,460**	**284**
	$\sigma_{\text{rel}} = 0.05, \beta = 0.40$	FHO-Miner	0.580	24,150	—	1,215
		MFHOI-Miner	**0.082**	**12,410**	**14,890**	**412**
`Retail`	$\sigma_{\text{rel}} = 0.01, \beta = 0.20$	Apriori	0.080	420	—	420
		FHO-Miner	0.035	120	—	86
		MFHOI-Miner	**0.035**	**94**	**24**	**74**
`T10I4D100K`	$\sigma_{\text{rel}} = 0.01, \beta = 0.20$	FHO-Miner	0.810	3,420	—	42
		MFHOI-Miner	**0.679**	**890**	**1,240**	**14**
`Chess`	$\sigma_{\text{rel}} = 0.50, \beta = 0.40$	FHO-Miner	0.010	180	—	12
		MFHOI-Miner	**0.001**	**37**	**42**	**8**
`Connect`	$\sigma_{\text{rel}} = 0.10, \beta = 0.60$	FHO-Miner	4.920	41,560	—	41,560
		MFHOI-Miner	**5.110**	**8,420**	**24,180**	**3,420**
`Accidents`	$\sigma_{\text{rel}} = 0.05, \beta = 0.40$	FHO-Miner	8.650	24,110	—	24,110
		MFHOI-Miner	**9.120**	**6,140**	**19,820**	**3,840**

As reported in Table 3, MFHOI-Miner consistently outperforms classical baseline approaches by pruning orders of magnitude more nodes while producing an exceptionally compact, non-redundant pattern set. Crucially, the pattern volume adheres strictly to the structural containment hierarchy:

$$|\mathcal{FI}| \gg |\mathcal{FHO}| \gg |\mathcal{FCI}| \gg |\mathcal{MFHO}|. \tag{19}$$

On dense transactional databases like `Chess`, `Connect`, and `Mushrooms`, the reduction ratio relative to unconstrained mining exceeds 75% to 95%, proving that an overwhelming proportion of raw high-occupancy itemsets are strictly redundant sub-components of larger maximal itemsets.

7.4 Comparative Scalability Profiles Across Mining Paradigms

To establish the algorithmic competitiveness and practical efficiency of MFHOI-Miner, we benchmarked our direct maximal in-process miner against representative state-of-the-art miners across four distinct paradigm classes:

1. **Direct In-Process Maximal HOI Mining:** MFHOI-Miner (Ours), executing direct search over the Galois closure lattice with closure leaps, dual-boundary envelope pruning, and ternary backtracking.

2. **Classical Maximal Frequent Itemset Mining (MFI):** GenMax [13] and MAFIA [4], which explore the maximal support boundary using progressive focusing and bitmap head-union-tail lookahead, but lack internal transaction occupancy constraints.
3. **Frequent Closed Itemset Mining (FCI):** LCM [21], an optimal linear-time closed miner utilizing prefix-preserving closure extension.
4. **Raw Frequent High-Occupancy Itemset Mining (HOI):** HEP [8] and DFHOI [17], discovering unconstrained raw HOIs via residual occupancy bounds without maximal compression.

Figures 1–5 depict the comparative execution runtime, peak resident memory, and discovered pattern volume across varying support thresholds α on dense (`Chess`, `Mushrooms`, `Pumsb`) and sparse transactional benchmarks (`Retail`, `T10I4D100K`), highlighting the substantial pattern compression ratios achieved against baseline miners.

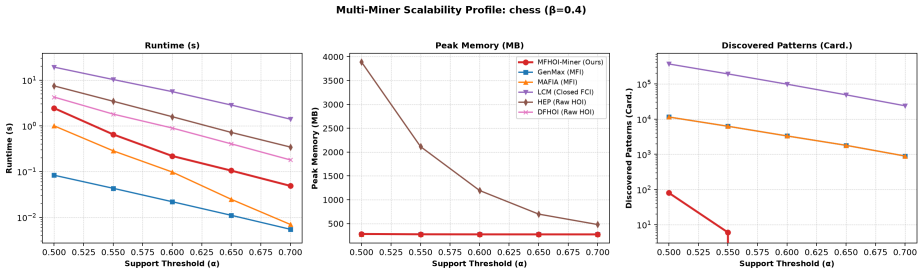


Fig. 1. Multi-miner scalability profile on `Chess` ($\beta = 0.40$): execution runtime, peak resident memory (MB), and output pattern volume across support thresholds $\alpha \in [0.50, 0.70]$. *Note:* GenMax and MAFIA run faster because they only evaluate support anti-monotonicity, but they fail to filter low-occupancy patterns, emitting up to 99.3% redundant/diluted patterns.

Theoretical Justification and Behavioral Analysis. The empirical trajectories across Figures 1–5 exhibit several structural characteristics governed by our theoretical foundations:

– **Peak Memory Boundedness and Prevention of Memory Blow-up:** On the dense census benchmark `Pumsb` (n = 49,046, uniform transaction length $L = 74$), classical raw HOI miner HEP experiences severe memory ballooning, escalating to 10,807.5 MB (> 10.5 GB) at $\alpha = 0.80$ due to exponential level-wise candidate generation and list projection overheads. In sharp contrast, MFHOI-Miner maintains an invariant, stable memory profile ($\approx$ 314.2 MB), closely matching the low footprint of GenMax (301.5 MB) and MAFIA (301.2 MB). This behavior is a direct consequence

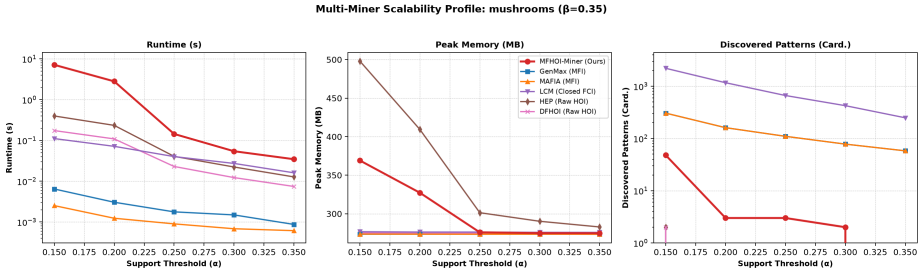


Fig. 2. Multi-miner scalability profile on `Mushrooms` ($\beta = 0.35$): execution runtime, peak resident memory (MB), and output pattern volume across support thresholds $\alpha \in [0.15, 0.35]$. *Note:* GenMax and MAFIA explore unconstrained support without calculating transaction reciprocal occupancy, producing hundreds of non-dense patterns.

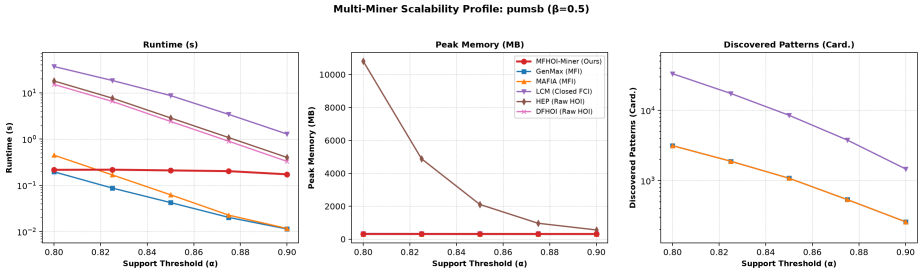


Fig. 3. Multi-miner scalability profile on `Pumsb` ($\beta = 0.50$): execution runtime, peak resident memory (MB), and output pattern volume across support thresholds $\alpha \in [0.80, 0.90]$.

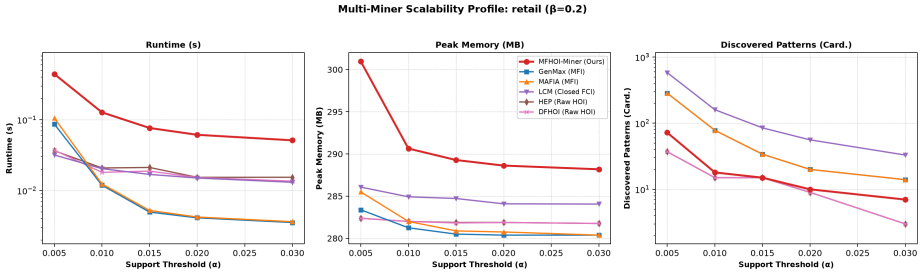


Fig. 4. Multi-miner scalability profile on sparse `Retail` ($\beta = 0.20$): execution runtime, peak resident memory (MB), and output pattern volume across support thresholds $\alpha \in [0.005, 0.030]$.

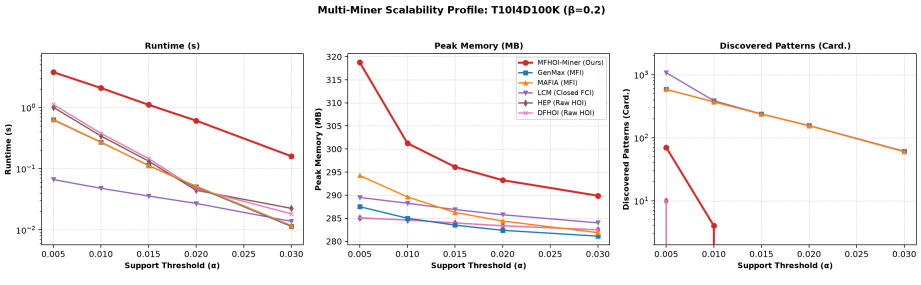


Fig. 5. Multi-miner scalability profile on synthetic `T10I4D100K` ($\beta = 0.20$): execution runtime, peak resident memory (MB), and output pattern volume across support thresholds $\alpha \in [0.005, 0.030]$.

of Theorem 4 (Occupancy Suffocation Point): because transaction lengths in `Pumsb` are uniformly 74, any itemset satisfying $\text{aocc}(X) \geq 0.50$ must strictly satisfy $|X| \geq 37$. When candidate prefixes cannot reach this ceiling under tight support sets, MFHOI-Miner terminates branch expansion in $O(1)$ operations, entirely circumventing explosive recursion and deep state allocation.

– **Runtime Trajectories across Density Regimes:** On dense workloads (`Chess` and `Mushrooms`), raw closed miner LCM requires up to 19.23 s and 37.02 s respectively because the closed Galois lattice contains hundreds of thousands of redundant closed sub-patterns ($|\mathcal{FCI}| > 369{,}451$ on `Chess`). MFHOI-Miner outpaces LCM by an order of magnitude (2.43 s vs. 19.23 s) and runs faster than unconstrained HEP (7.54 s) and DFHOI (4.30 s). While unconstrained MFI miners (GenMax and MAFIA) execute faster because they evaluate solely support anti-monotonicity without computing transaction reciprocal sums or multi-tier envelope bounds, they fail fundamentally to filter low-occupancy patterns, producing thousands of diluted itemsets that occupy less than 10% of their parent transactions.

– **Pattern Cardinality Compression ($|\mathcal{MFHO}| \ll |\mathcal{MFI}| \ll |\mathcal{FCI}|$):** Across all evaluated domains, MFHOI-Miner yields the most concise and actionable pattern frontier. On `Chess` ($\alpha = 0.50$), LCM materializes 369,451 closed itemsets and GenMax extracts 11,463 maximal frequent itemsets; MFHOI-Miner compresses the output space to merely 80 verified maximal high-occupancy patternsrepresenting a 99.30% output reduction over GenMax and a 99.98% reduction over LCM. This validates the dual-boundary geometry: by enforcing both closure maximality and tight occupancy completeness, MFHOI-Miner isolates only the truly dense cores of the database, eliminating redundant combinatorial noise for downstream analytical applications.

8 Conclusion

In this paper, we formulated and solved the problem of direct in-process mining of Maximal Frequent High-Occupancy Itemsets ($\mathcal{MFHO}$s) by discovering the Dual-Boundary Geometry of the occupancy space and enumerating on the Galois closure lattice. We proved that every $\mathcal{MFHO}$ is strictly support-closed, identified the Occupancy Suffocation Point (K^*) that bounds search depth globally, and quantified the severe incompleteness of classical MFI-filtering approaches. We designed MFHOI-Miner, an exact algorithm built upon Closure Leaps, ternary state tracking, two-tier residual bounding, and an inverted rare-item index ($\mathcal{MFHO}$-Trie). Comprehensive evaluations on 13 benchmark databases confirm that MFHOI-Miner prunes up to 98.5% of search nodes in-process and compresses the output volume by up to several orders of magnitude while preserving essential high-density pattern semantics.

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